

MODEL QUESTION 2026

BSCSIT 4th Semester, 2026

Course Title: Computer Graphics

Group A

Very short questions (5 x 2 = 10)

1. Resolution and aspect ratio
2. Blobby Objects
3. Orthographic Projection
4. Virtual Reality
5. Polygon table

Group B

Short Questions. Attempt any SIX questions: (6 x 5 = 30)

1. Differences between computer graphics and image processing. List pros and cons of computer graphics.
2. Explain the Bezier curve and B-Spline curve.
3. Describe the 3D viewing pipeline with appropriate diagrams.
4. Explain any two functions in OpenGL. Write the basic commands to draw the pixel and polygon in OpenGL.
5. Define intensity attenuation. Distinguish between Gouraud shading and Phong shading model.
6. Calculate the total memory required to store a 12-minute video in a SVGA system with 24-bit true color and 30 fps.
7. What do you understand by solid modeling? Explain binary space partition method.

Group C

Long Questions. Attempt any TWO questions: (2 x 10 = 20)

1. Explain the working details of Mid-point circle algorithm? Trace the points along the line path having two end points (2,3) and (10,8) using Bresenham's line drawing algorithm.
2. Differentiate between object space and image space methods of hidden surface removal. Describe the Z-buffer hidden surface removal algorithm.
3. Why is the Liang Barsky Line Clipping Algorithm more efficient than Cohen Sutherland Algorithm? Explain the clipping procedure of Liang Barsky algorithm with suitable example.